CURRENT RESEARCH / COMPUTATIONAL PROTEIN DESIGN

AI-designed bispecific binders for Alzheimer's disease

Aβ42 targeting × transferrin receptor 1 (TfR1) | Goal: receptor-mediated delivery across the BBB

PRE-FILTER EXPLORATION (reported trajectories): Aβ42 1,342 | TfR1 / BBB arm 2,051

01 TARGET PREPARATION
Structure-guided epitopes
Aβ42: 9CO4 • TfR1: 6WRV

02 DE NOVO DESIGN
BindCraft + ProteinMPNN
Generate, redesign, AF2-score

03 FILTER & RANK
Stability + developability
23 Aβ42 / 224 TfR1 retained

04 FUSION LIBRARY
5 Aβ42 × 5 TfR1 binders
5 linkers × 2 domain orders

05 FOLD & SCREEN
AlphaFold2 / ColabFold
250 fusion predictions scored

442
accepted arm designs
62 Aβ42 + 380 TfR1 (reported)

250
fusion designs modeled
25 binder pairs × 10 variants

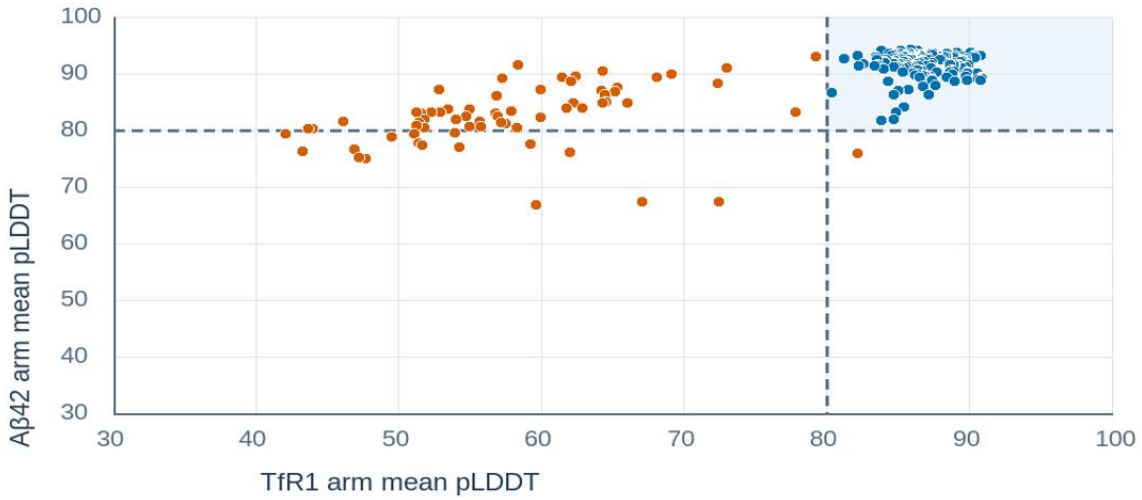
183 / 250
pass Stage 8 gates
73.2% computational pass rate

FUSION-LEVEL COMPARISON • All 250 designs; one top-ranked prediction per design

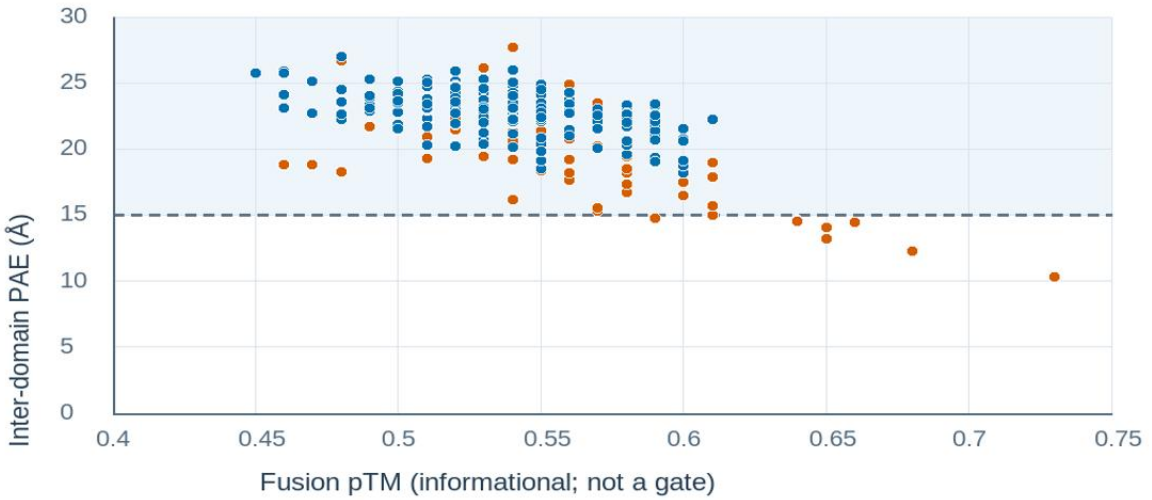
● Pass all gates (n = 183) ● Fail ≥1 gate (n = 67)

Dashed lines: required thresholds

Per-arm folding confidence across fusion designs



Whole-chain confidence vs. domain independence



Gate: both arms pLDDT >80 AND inter-domain PAE >15 Å. High inter-domain PAE reflects uncertain relative orientation.